IEG313-Quantum Computing for Engineers: Theory and Applications

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Outline

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2. Quantum Registers
3. Quantum Parallelism
4. Quantum Speed up
5. Quantum Decoherence
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Vector Space

A vector space (also called a linear space) is a mathematical structure formed by a collection of objects called vectors, which can be added together and multiplied by scalars (elements from a field like real or complex numbers). These operations must satisfy a specific set of rules known as vector space axioms.

<u>Components:</u>

- Set of vectors: Denoted by **V**
- Field of scalars: Denoted by **F** (e.g., $\mathbb{R}, \mathbb{C}$)
- Operations:
  - Vector addition: $u + v \in \boldsymbol{V}$
  - Scalar multiplication: $a \cdot v \in \boldsymbol{V}, where\ a \in \boldsymbol{F}$

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Axioms of a Vector Space

For all vectors $u, v, w \in \boldsymbol{V}$ and scalars $a, b \in \boldsymbol{F}$, the following must hold:

- Closure under addition: $u + v \in \boldsymbol{V}$
- Commutativity of addition: $u + v = v + u$
- Associativity of addition:
$$(u + v) + w = u + (v + w)$$
- Existence of additive identity: There exists $0 \in \boldsymbol{V}$ such that $v + 0 = v$ for all $v \in \boldsymbol{V}$
- Existence of additive inverse: For each $v \in \boldsymbol{V}$, there exists $-v \in \boldsymbol{V}$ such that $v + (-v) = 0$
- Closure under scalar multiplication:
$$a \cdot v \in \boldsymbol{V}$$
- Multiplicative identity: $1 \cdot v = v$ for all $v \in \boldsymbol{V}$
- Associativity of scalar multiplication:
$$a(bv) = (ab)v$$
- Distributivity over vector addition:
$$a(u + v) = au + av$$
- Distributivity over scalar addition:
$$(a + b)v = av + bv$$

Inner Product Space

- An inner product space is a special kind of vector space that allows you to measure angles, lengths, and orthogonality—essentially bringing geometry into algebra.

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Inner Product Space

Definition: An inner product space is a vector space ($\boldsymbol{V}$) over a field ($\boldsymbol{F}$) (usually ($\mathbb{R}$) or ($\mathbb{C}$)) equipped with an inner product: a function $[\langle \cdot, \cdot \rangle : \boldsymbol{V} \times \boldsymbol{V} \rightarrow \boldsymbol{F}]$ that satisfies the following properties for all vectors ($u, v, w \in \boldsymbol{V}$) and scalars ($\alpha \in \boldsymbol{F}$)

Inner Product Axioms:

1. **Conjugate Symmetry:** $[\langle u, v \rangle = \overline{\langle v, u \rangle}]$ (For real spaces, this just means symmetry: ($\langle u, v \rangle = \langle v, u \rangle$))
2. **Linearity in the First Argument**: $[\langle \alpha u + w, v \rangle = \alpha \langle u, v \rangle + \langle w, v \rangle]$
3. **Positive-Definiteness**: $[\langle v, v \rangle \geq 0 \; and \; \langle v, v \rangle = 0 \Leftrightarrow v = 0]$

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Inner Product Space

- The inner product helps to define:-
- **Length (norm)**: ($|v| = \sqrt{\langle v, v \rangle}$)
- **Angle between vectors**: via cosine similarity
- **Orthogonality**: ($\langle u, v \rangle = 0$) means (u) and (v) are perpendicular

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Examples of Inner Product Spaces	Inner Product
($\mathbb{R}^n$)	($\langle x, y \rangle = \sum x_i \, y_i$) (dot product)
($\mathbb{C}^n$)	($\langle x, y \rangle = \sum x_i \, \overline{y_i}$)
Function space ($C[0,1]$)	($\langle f, g \rangle = \int_0^1 f(t)g(t)dt$)

Hilbert Space

- A Hilbert space is a complete vector space equipped with an inner product.
- Formally, it satisfies:

1. Vector space structure
2. Inner product: For any two vectors $|u\rangle$ and $|v\rangle$, there is an inner product $\langle u|v\rangle$ that gives a complex number, allowing us to define notions like length and angle.
3. Completeness: Every Cauchy sequence of vectors converges to a vector within the space.

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Hilbert Space

- Mathematically, this is called a Cauchy sequence: for every very small number $\epsilon>0$, there is a point in the sequence after which all vectors are within ϵ of each other.
- Completeness means: If such a "getting closer and closer" sequence exists, it must actually converge to a vector that is also in the space.

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Hilbert Space

- Completeness guarantees that: sequences of quantum states that get closer and closer will have a well-defined state as their limit.so calculations and approximations always stay inside the allowed state space.
- Typical examples include:

1. $\mathbb{C}^n$ (finite-dimensional complex vector spaces)
2. Spaces of square-integrable functions L^2

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Quantum Registers

- A quantum register is where quantum computers store and manipulate quantum data.
- A quantum register is a collection of qubits (quantum bits) grouped together.
- While a single qubit can be in a state like $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, a register of n qubits can be in a superposition of 2^n states at once.
- It's like the quantum version of a computer's memory register, but it can hold much more information in parallel because of superposition.

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Quantum Parallelism

- One instruction can affect many possible states at once, giving the sense of doing computations “in parallel.”
- Quantum parallelism is the ability of a quantum computer to process many inputs at the same time thanks to superposition.
- If you apply an operation (a quantum gate) to a register in superposition, the operation acts on all parts of the superposition simultaneously.

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Quantum Speed up

- Quantum computers can use quantum effects to reduce computation time for certain problems.
- Quantum speedup means that for some problems, a quantum algorithm can solve them much faster than the best-known classical algorithm.
- It doesn't mean every task is faster, only specific types—like factoring large numbers (Shor's algorithm) or searching unsorted data (Grover's algorithm).

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Quantum Decoherence

- Decoherence is like a quantum system “forgetting” it was quantum because of unwanted environmental interference.
- Quantum decoherence happens when a quantum system interacts with its surroundings and loses its quantum behaviour (superposition and entanglement).
- The system starts to behave classically instead of quantum-mechanically.

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Quantum Noise

- Quantum noise is the main cause of errors in quantum computers and is closely related to decoherence.
- Quantum noise refers to the random errors or disturbances that affect qubits during computation or storage.
- It can come from heat, electromagnetic radiation, or imperfections in the quantum hardware.

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THANK YOU!